Final Project: Database-Backed Application

Overview

Design and implement a database project demonstrating proper database design, SQL implementation, and data management principles. Project scope varies by track.

General Requirements (All Tracks)

Database Design

- Complete ER Diagram (entities, relationships, cardinalities, PKs/FKs)
- Normalized to 3NF minimum

GitHub Repository

- Public repository with incremental commits (minimum 10)
- Include repository link in submission

Database

- Oracle (use provided credentials)

Required SQL Files

- `sql/schema.sql` - DDL with CREATE TABLE statements, PKs/FKs/constraints
- `sql/sample_data.sql` - INSERT statements with representative data
- `sql/reset.sql` - DROP/TRUNCATE statements to reset database

Guidelines

- Use AI assistants but understand all generated code
- Must explain design decisions in the report
- Generate a proper report with Summary, Introduction, Data modeling, Design decisions, screenshots of database design with one line of explanation for each

- Choose any project of your choice

Project Ideas

1. Donation & Fundraising Platform

2. Learning Management System

3. Property Rental Management

4. Restaurant Order Management

5. Product Review & Rating Platform

6. Event Management Platform

7. Help Desk Ticket System

8. Job Board & Career Portal